

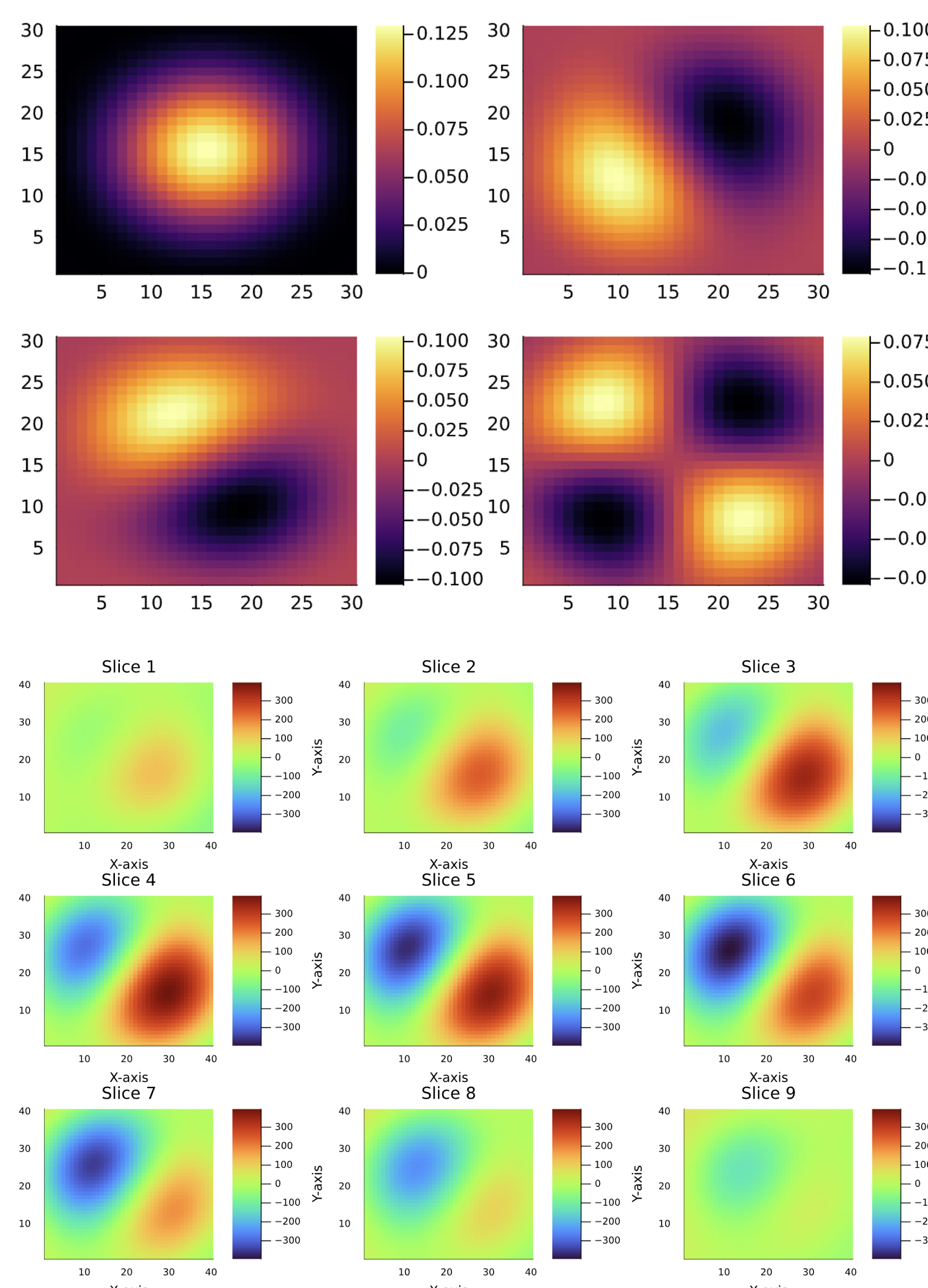
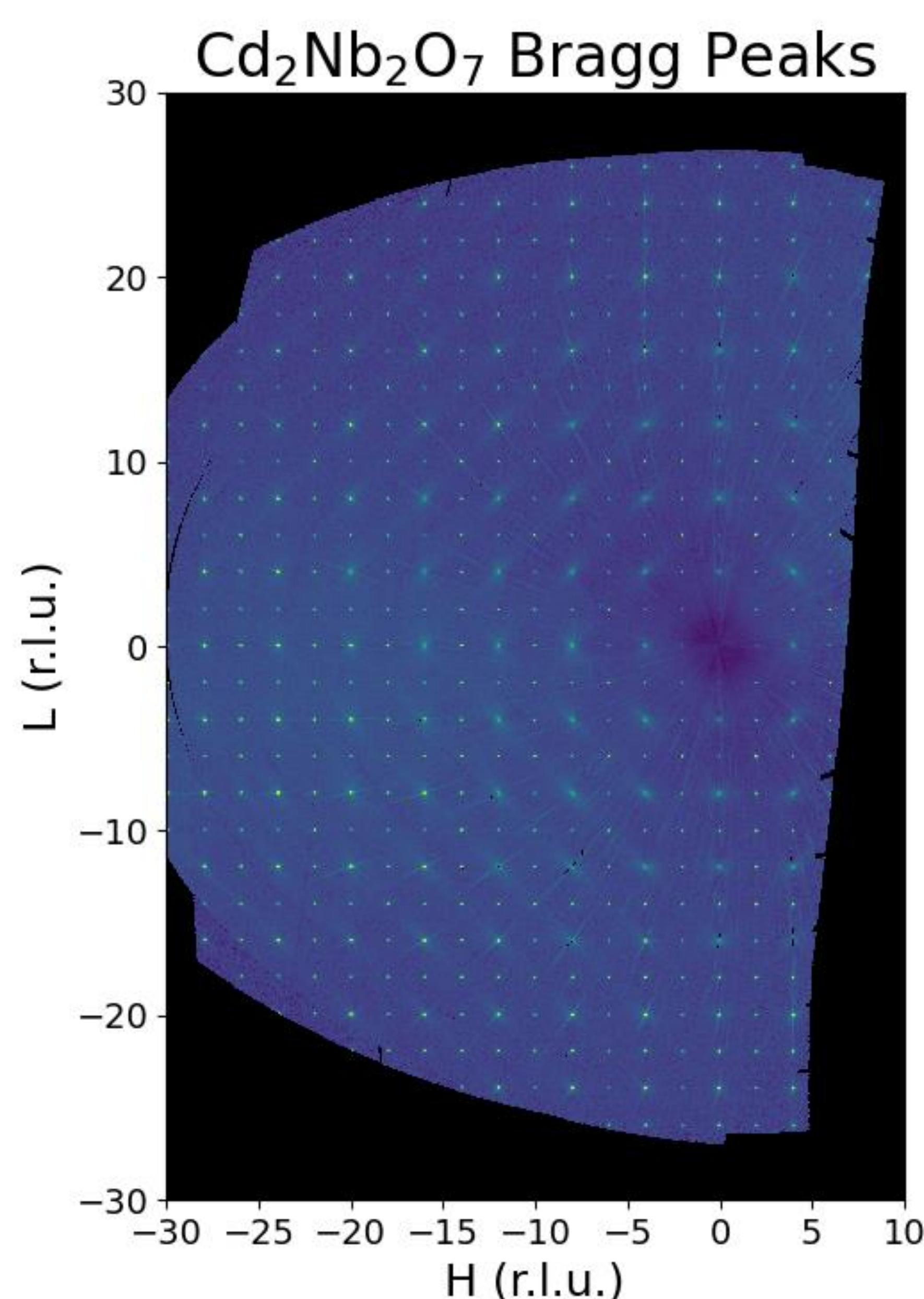
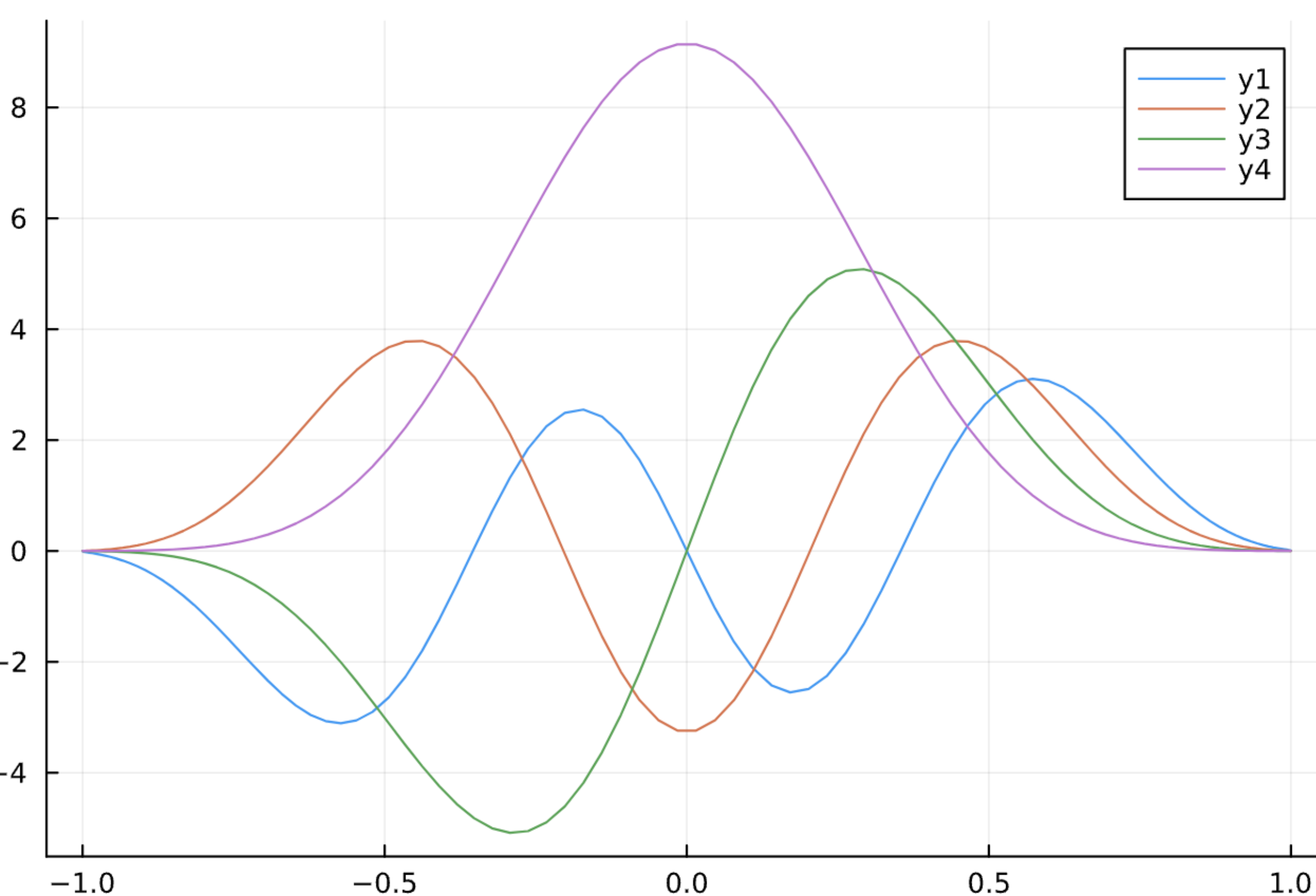
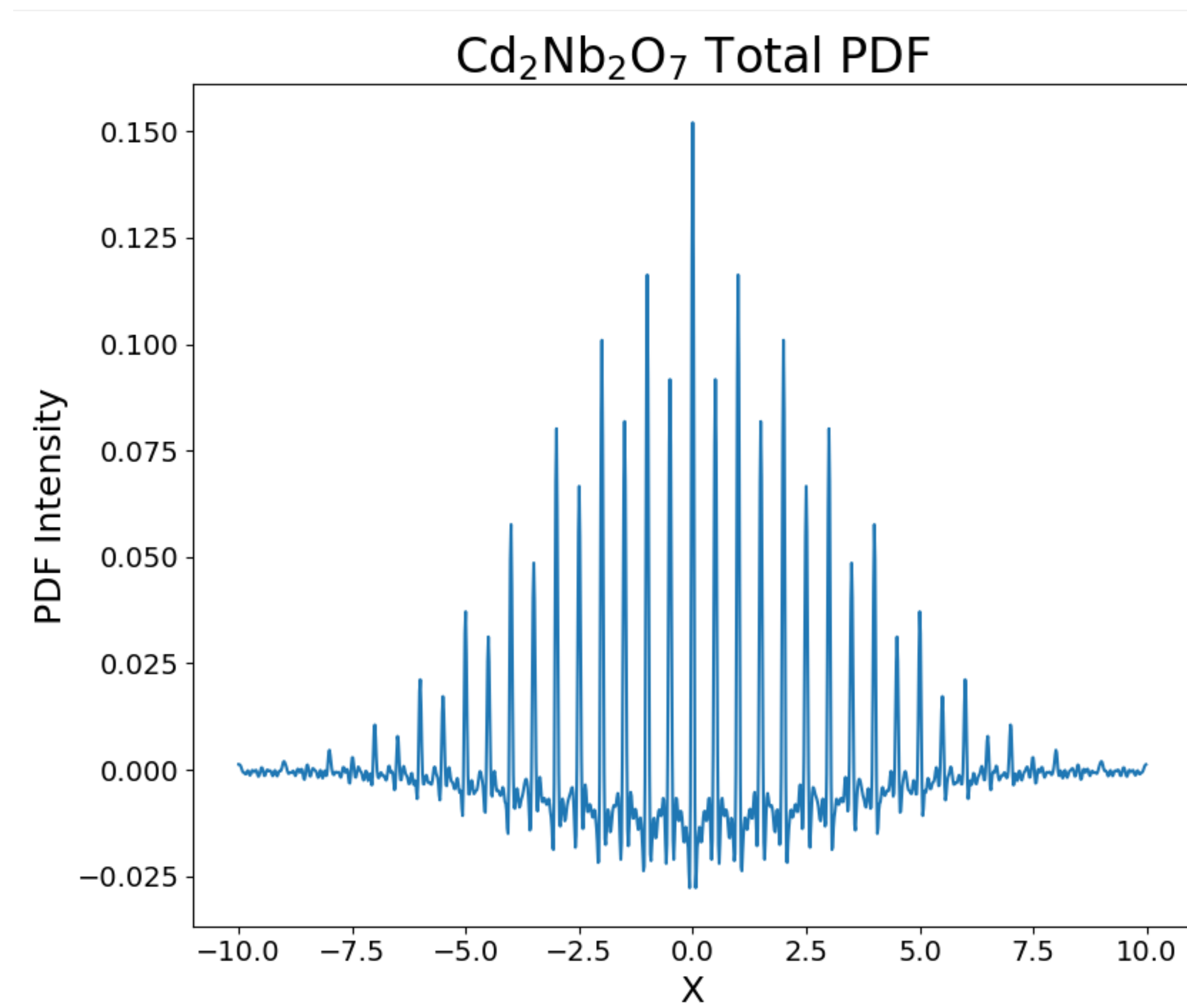
SPECTRAL ANALYSIS OF MATERIAL STRUCTURES

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MOTIVATION

- The "punch-and-fill" algorithm (PFA) used to generate 3D- Δ PDF maps can introduce unintended ripples or noise due to sharp edges in data interpolation.
- To mitigate this noise, we propose leveraging the inherent sparsity of these maps; the PDF is essentially zero except for discrete features representing pairwise interatomic vector probabilities.



SIGNIFICANCE

We use functions localized both in their Fourier transform and in real space, to minimize leakage artifacts in PDFs. Using a sparse representation for the diffuse scattering in the real space, we create interpretable PDFs with minimal noise. We employ Nystrom method and Discrete Prolate Spheroidal Sequences (Slepian 1978) and solve Fredholm equations (Numerical Recipes)

IMPACT

By minimizing noise and artifacts in Pair Distribution Functions (PDFs), our method provides more accurate and interpretable data for materials science research. This improvement allows for Enhanced Material Characterization, Improved Data Quality, Broader Applicability and optimized computational efficiency.

MAJOR ACCOMPLISHMENTS

- Discrete Prolate Spheroidal Sequences (DPSS), are optimal sequences for spectral concentration problems and are extensively used in signal processing and data analysis. Utilizing and modifying the DPSS equations (Simons and Wang 2011), and solving the Fredholm equations (Numerical Recipes) an algorithm for spectral analysis in 1, 2 and 3 Dimensions was implemented.

$$D(x, x') = \frac{\sin[TW(x-x')]}{\pi(x-x')} \text{ for 1D, and}$$

$$D(x-x') = \frac{KJ_1(K||x-x'||)}{2\pi||x-x'||} \text{ for 2D}$$

- This was done in Julia, a language known for its speed in numerical analysis. Stepping through the increasing dimensions, 2x-4x speedups were achieved using Multithreading SIMD and Parallel Programming methods

NEXT STEPS

- We usually select the punch size manually.
- Sometimes there are weaker results because of these choices.
- We can use Nonlinear Nonconvex-OPTimization problems (NOPT) as an extension for the compressed sensing problem
- These can optimize for the punch size.

- Slepian, D. (1978). *Prolate Spheroidal Wave Functions, Fourier Analysis, and Uncertainty—V: The Discrete Case*. *Bell System Technical Journal*, 57(5), 1371-1430. <https://doi.org/10.1002/j.1538-7305.1978.tb02104.x>
- Press, W. H., Teukolsky, S. A., Vetterling, W. T., & Flannery, B. P. (2007). *Numerical recipes: The art of scientific computing (3rd ed.)*. Cambridge University Press.

*Safety tip: Maintain good posture, take regular breaks, keep your workspace ergonomically optimized and clean, ensure proper lighting and electrical safety, and manage stress for overall well-being.



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